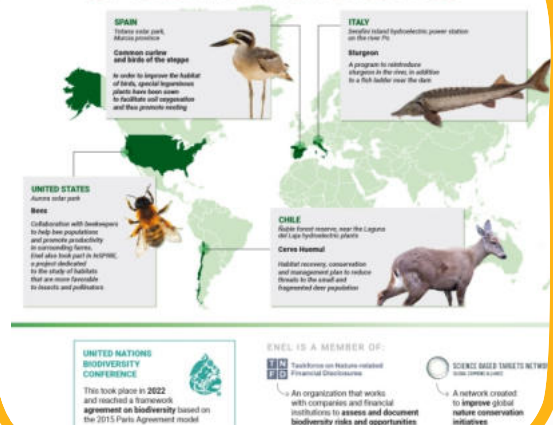
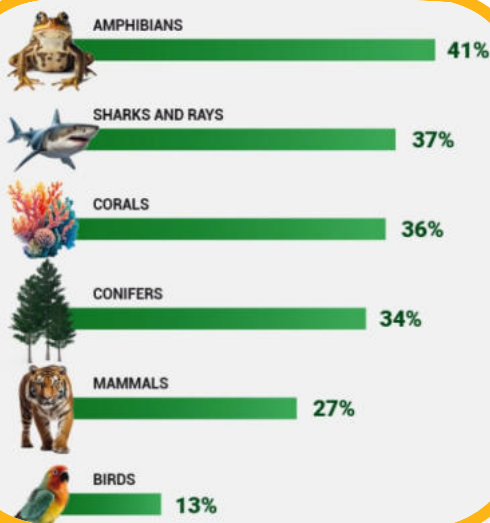




BIODIVERSITY



DEFINITIONS**BIODIVERSITY:**

The biodiversity is the degree of variation within or among the species exists on different region of the earth.

BIOLOGICAL CLASSIFICATION:

The scientific procedure of arranging organisms into a series of groups sub groups on the basis of their similarities and dissimilarities.

HOMOLOGOUS:

Organs which are similar in structure but different in function are known as homologous.

ANALOGOUS:

Organs which are different in structure but similar in function are known as analogous.

TAXANOMIC HIERARACHY:

The taxa are arranged in ascending order and form a ladder called taxonomic hierarchy.

DEFORSTATION:

The cutting down of trees for the conversion of forest into non- forest land is known as deforestation.

SPECIES:

It is a group of organisms capable of interbreeding and produce fertile offspring.

MYCELIUM:

Body of multicellular eukaryotic fungi is called mycelium

HYPAE:

Cell wall of multicellular eukaryotic fungi is made up of thread like structure called hyphae.

SHORT/DETAILED ANSWER & QUESTIONS**Q. What is biodiversity? Explain distribution and importance of biodiversity?**

Ans. **BIODIVERSITY**

DEFINITION:

"The biodiversity or biological diversity is the degree of variation within or among the species exist on different regions of the earth"

DISTRIBUTION OF BIODIVERSITY

Biodiversity is not distributed evenly on earth.

1. TROPICAL REGIONS:

Biodiversity is richest in the tropic's region.

2. TEMPERATE REGIONS:

Temperate regions also have many species.

3. POLAR REGIONS:

Few species are present in polar regions. Biodiversity found on earth today is the result of 4 billion years of evolution. All life consisted of bacteria and similar unicellular organisms.

IMPORTANCE OF BIODIVERSITY:**SOURCE OF FOOD:**

Biodiversity provides food for humans.

SOURCE OF DRUGS:

A significant proportion of drugs are derived directly or indirectly from biological sources.

SOURCE OF INDUSTRIAL MATERIALS:

A wide of range industrial materials e.g., building materials, fibers, dyes, resins, gums, adhesive, rubber, and oil are derived directly from biological resources.

MAINTAINANCE OF ECOSYSTEM:

Biodiversity plays an important role in making and maintaining ecosystems. It plays a in regulating the chemistry of our atmosphere and water supply.

Q. What is classification? Describe the aims and basis of classification.

Ans: **CLASSIFICATION:**

DEFINITION:

The arrangement of organisms into groups and sub-groups on the basis of similarities and dissimilarities is called classification.

KNOWN NUMBER OF ANIMALS AND PLANTS SPECIES:

Over 1.5 million types of animals and over 0.5 types of plants are known to biologist.

SIZE:

The range in complexity from small and simple bacteria to large and complex human beings

HABITAT:

Some of the organisms live in sea, some on land, some walk, others fly and still others are stationary.

NEED FOR CLASSIFICATION:

Each species has its own life. When there are so many kinds of organism, biologist classify the organisms to study such a large collection in any easy way.

TAXONOMY:

The branch of Biology which deals with scientific naming and classification of organisms is called taxonomy.

SYSTEMATICS:

The branch which deals with classification and also traces the evolutionary history of organisms is known as systematics.

AIMS OF CLASSIFICATION:

The main aim of classifications is

1. To determine similarities and differences among organisms so that they can be studied easily.
2. To find the evolutionary relationships among organisms.

BASIS OF CLASSIFICATION:

Classification is based on relationship among organisms and such relationship is got through similarities in the form or structure.

These similarities are seen in

1. Structures both internal and external
2. Biochemistry
3. Modern genetics

The similarities suggest that all organisms are related to one another at some point in their evolutionary histories. However, some organisms are more closely related than others.

FOR EXAMPLE:

Sparrows are more closely related to pigeons than to insects. It means sparrows and pigeons have common evolutionary histories.

Q. What is taxonomic hierarchy?

Ans: **TAXA**

Definition:

The groups in which organisms are classified are known as taxonomic categories or taxa.

TAXONOMIC HIERARCHY

Definition:

The taxa are form a ladder called Taxonomic hierarchy. All the organisms are divided into five kingdoms. So, kingdom is the largest taxon, or the basis of similarities, each kingdom is further divided into smaller taxa in the following ways:

Phylum: A Phylum is a group of related classes.

Class: A class is a group of related order.

Order: An order is a group of related families.

Family: A family is a group of related genera.

Genus: A genus is a group of related species.

Species: A species consists of similar organisms.

SIMPLE CLASSIFICATION OF TWO ORGANISMS

TAXA	HUMAN	PEA
Kingdom	Animalia	Plantae
Phylum	Chordate	Magnoliophyte
Class	Mammalia	Magnoliopsida
Order	Primates	Fabales
Family	Moninidae	Fabaceae
Genus	Homo	Pisum
Species	Sapiens	Sativum
Scientific Name	Homosapiens	Pisum Sativum

Q. What is contribution of Aristotle in classification of living organisms?

Ans. The earliest known system of classification of organisms comes from the Greek philosopher Aristotle. He classified all living organisms known at that time in two groups.

- Kingdom Plantae
- Kingdom Animalia

Q. Why two kingdom classification system was rejected?

Ans. Some taxonomists found this system unworkable because:

- Many unicellular organisms like Euglena have both plant like (Presences of chlorophyll and animal like heterotrophies mode of nutrition and lack of cell wall) characters. So, there should be a separate kingdom for such organisms.
- This system also ignores the difference between organisms having prokaryotic and those having eukaryotic cells.

Q. Who proposed three-kingdom classification and what were the problems with Kingdom system?

Ans. **THREE KINGDOM CLASSIFICATION:**

In 1886 Ernst Haeckel solved the first objection and presented three kingdom classification system.

OBJECTIONS:

• This system does not clear the difference between prokaryotic and eukaryotic. **Fungi** were placed in Kingdom plantae. Some biologist disagreed about the position of fungi in **Kingdom plantae**. Fungi resemble plants in many ways but are not autotrophs. They are special form of heterotrophs and get their food by absorption. They do not have cellulose in their cell walls, rather they possess chitin.

Q. Describe the four-kingdom classification?

Ans. **FOUR KINGDOM CLASSIFICATION:**

After the clear concept for kingdom Protista, Copeland (1959) came forward with a four kingdom system to classify the living beings. He designed a new kingdom named as Monera to place all the lower protists which include prokaryotic unicellular organisms and remaining single called eukaryotic organisms were include in Protista.

Q. What is the role of E- Chatton?

Ans. In 1937, E-chatton suggested the terms of "Prokaryote" to describe bacteria and "Eukaryote" to describe animal and plant cells.

Q. What is the basis of five kingdom classification?

Ans. The system of classification was based on.

CELLULAR ORGANIZATION:

The levels of cellular organization, i.e., Prokaryotic, unicellular eukaryotic and multicellular eukaryotic.

MODES OF NUTRITION:

The principal mode of nutrition i.e., photosynthesis, absorption and ingestion.

Q. How can you divide five kingdoms into two groups on the basis of types of cells?

Ans. Five kingdoms can divide into two groups on the basis of cells in following ways.

1. Kingdom Monera (organisms with Prokaryotic cells)
2. Kingdom Protista, Fungi, Plantae, Animalia, (organism with eukaryotic cells)

Q. What are prions and viroid's?

Ans. Prions and viroids are acellular forms of organisms. Both of these particles cause infectious diseases in certain plants. They are not included in five kingdom classification.

PRIONS:

They are composed to proteins only.

VIROIDS:

They are composed of circular RNA only.

Q. Classify that virus are living or non-living?

Ans. Viruses are at the borderline of living and non-living. They have both living and non living features which distinguish their unique Identity.

NON-LIVING FEATURES:

- Due to their crystalline nature, they are considered as non-living.
- They are acellular i.e., that do not have a cellular organization.

LIVING FEATURES:

- They contain DNA or RNA normally enclosed in a protein coat.
- They reproduce, but only in living cells.
- They cause a number of diseases in living cells.

CONCLUSION:

They are not considered as organisms and thus are not assigned any place in the five-kingdom classification system.

Q. Define Binomial Nomenclature.

Ans. **INTRODUCTION:**

Binomial Nomenclature is the method of giving scientific names to living organisms.

MEANING:

"Bi-nomial" means "Two names"

FORMAT:

As the word "binomial" suggest, the scientific name of any living organisms consists of two names: the first is the **GENUS** name and the second one is the name of **SPECIES**.

EXAMPLE:

The scientific name of human beings is **HOMOSAPIENS**.

Q. Write down the common name and scientific name of some plants and animals.

PLANTS		ANIMALS	
Common Name	Biological Name	Common Name	Biological Name
Onion plant	Allium cepa	Frog	Rana tigrine
Mango plant	Mangifera indica L.	Cat	Felis catus
Neem plant	Azadirachta indica	Housefly	Musca domestica
Potato	Solanum tuberosum	Man	Homo sapiens
Matar	Pisum sativum	Dog	Canis lupus familiaris
China rose	Hibiscus rosa-sinensis	—	—

Q. What are the main causes of loss of biodiversity in Pakistan?

Ans. Pakistan today faces severe threats to its animal and plant species. The main causes of this loss are.

- Loss of natural habitats.
- Rapid growth in population.
- Prevailing poverty in rural areas.
- Low literacy rate.

Q. Name the national animal and birds of Pakistan.

Ans. **NATIONAL ANIMAL:**

Markhor is the national animal of Pakistan.

NATIONAL BIRD:

Chukor partridge is the national bird of Pakistan.

Q. Describe Five Kingdoms Classifications.

Ans. **KINGDOM:**

It includes eukaryotic unicellular and simple multicellular organisms. These organisms may be photosynthetic or heterotrophic.

Example:

KINGDOM MONERA:

It includes prokaryotic organisms i.e., they are made of prokaryotic cells. Monerans are unicellular, although some types form chains, clusters, or colonies of cells. Most are heterotrophic but some perform photosynthesis.

EXAMPLES:

- Bacteria.
- Cyanobacteria.

KINGDOM FUNGI:

• It includes eukaryotic multicellular heterotrophs which are absorptive in their nutritional mode.

• They live on organic material, secrete digestive enzymes and absorb small organic molecules.

Example:

- Mushrooms

KINGDOM PROTISTA:

• These are unicellular eukaryotic organisms, usually motile with a variety of feeding methods. Except yeast.

• Most protists are aquatic in nature.

• It includes protozoa and unicellular algae.

Example:

- Amoeba
- Paramecium
- Chlamydomonas.

KINGDOM PLANTAE:

• it includes eukaryotic multicellular autotrophs.

• They have sex organs and form embryo during life cycle

Example:

- Mosses, ferns and flowering
- plants are included in
- Kingdom plantae.

KINGDOM ANIMALIA:

• These are multicellular eukaryotes without cell wall and chloroplast in their cells.

• All animals including invertebrates and vertebrates with exception of protozoans are included in this kingdom.

• They all have heterotrophic ingestive mode of nutrition.

Example:

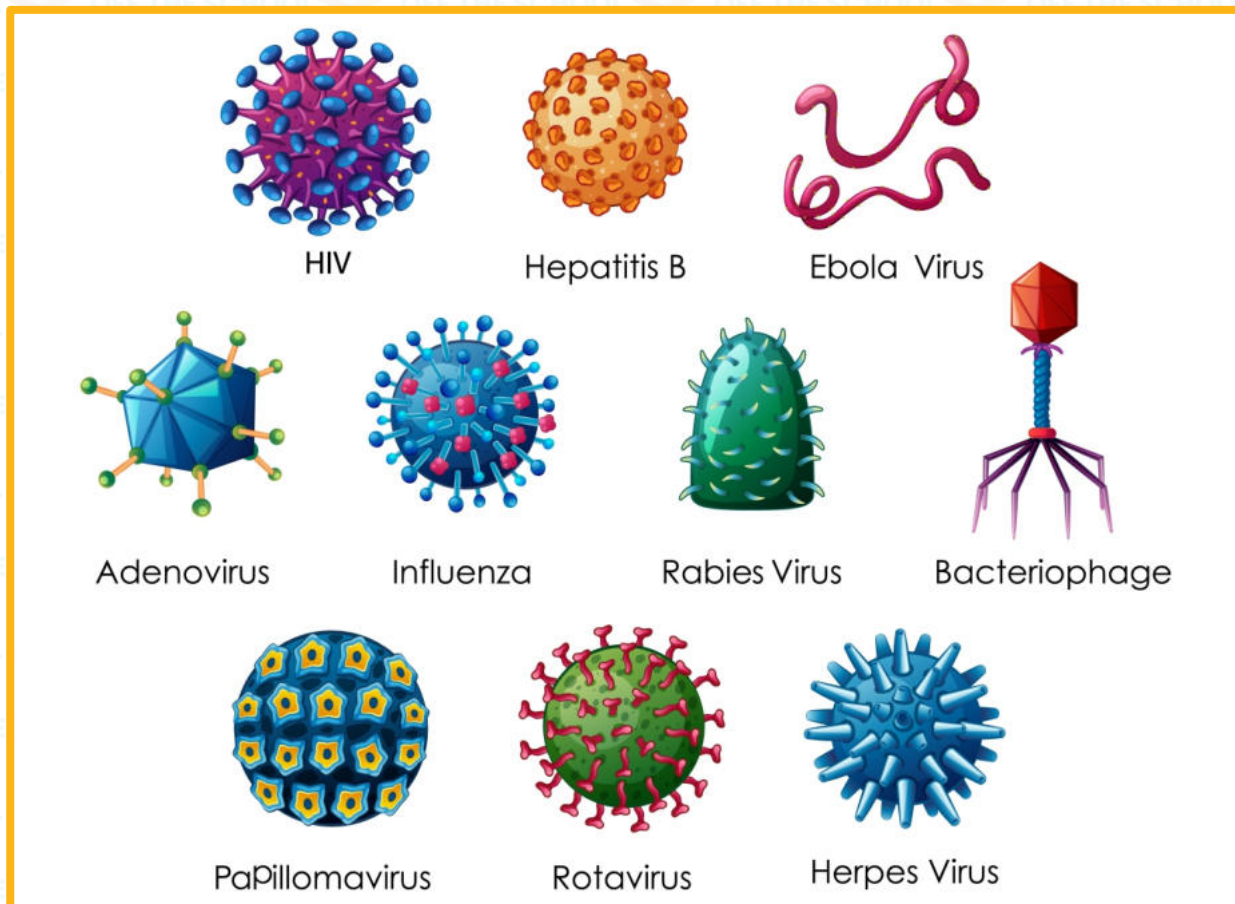
All animal including invertebrates and vertebrates are the members of this kingdom.

Q. What do you know about the Structure of virus?

Ans. **STRUCTURE OF VIRUS:**

They appear like a small rod, tadpoles or may be polyhedral or like little spheres. Viruses may consist of nucleic acid, capsids, envelopes and tail fibers. Their nucleic acid may consist of a single or several

molecules of DNA or RNA. The smallest viruses have only for genes while the largest have up to two hundred. The protein coat that encloses the nucleic acid is called a capsid. it may be of different shapes. Cape site is made up of protein subunits called capsomeres the number of capsomeres is characteristics of a particular virus. Some viruses have accessory structures called viral-envelopes that help them infect their hosts. They are membranes covering their capsids. The simplest viruses consist of single molecule of a nuclei acid (DNA or RNA) surrounded by a capsid, which is made up of different protein molecules.



Q.1. Why scientific naming is necessary?

Ans. Every recognized species on earth is given a two- part scientific name. This system is called "binomial nomenclature". These names are important because they allow people throughout the world to communicate unambiguously about animal species.

Q.2. How living organisms are classified into two kingdom classification?

Ans. In early days, organisms were classified into two major groups called animal and plant kingdom. The basis of two kingdom classification (divisions) was the presences or absence of cell- wall and chlorophyll.

Q.3. Why amoeba is not placed in animal kingdom?

Ans. Amoeba is not animal; however, they are classified in the protist kingdom. An amoeba's single cell appears to be not much more than cytoplasm held together by a flexible cell membrane.